

NexAfrica Machine Learning Internship

5-Week Project Guide: Customer Churn Prediction

Project Overview

Project Title: Customer Churn Prediction

Project Type: Binary Classification

The goal of this project is to build a Machine Learning model that predicts whether a customer is likely to leave a company's service (churn) or remain a customer.

This project simulates a real-world business problem commonly found in telecommunications, banking, insurance, subscription services, and e-commerce.

Project Workflow

Business Understanding → Data Collection → Data Cleaning → EDA → Feature Engineering → Model Training → Model Evaluation → Model Optimization → Feature Importance → Business Recommendations

WEEK 1: Business Understanding, Data Collection & Data Preparation

Objective

Understand the customer churn problem, explore the dataset, and prepare the data for analysis.

Task 1: Understand the Business Problem

Write a short project introduction answering:

- What is customer churn?
- Why is customer churn important to a business?
- What is the goal of the prediction model?
- How could predicting churn help the company?

Example Problem Statement

The objective of this project is to develop a Machine Learning classification model that predicts whether a customer is likely to churn based on demographic information, account details, subscription information, service usage, contract type, tenure, and billing information.

Task 2: Obtain and Load the Dataset

Use a **Customer Churn Dataset** from Kaggle.

Load the dataset using Pandas.

Inspect:

- Number of rows
- Number of columns
- Column names
- Data types
- Sample records
- Dataset structure

Task 3: Understand the Variables

Create a simple data dictionary.

Example:

Feature	Description	Type
CustomerID	Unique customer identifier	Categorical
Gender	Customer demographic information	Categorical
Tenure	Length of time as a customer	Numerical
Contract	Customer contract type	Categorical
MonthlyCharges	Monthly customer charge	Numerical
TotalCharges	Total amount charged	Numerical
PaymentMethod	Customer payment method	Categorical
Churn	Whether the customer left	Target

Identify:

- **Target variable:** Churn
- **Features:** All relevant variables used to predict churn

Task 4: Data Cleaning

Check for:

Missing Values

Identify and handle missing or blank values.

Duplicate Records

Check for duplicate customers or duplicate rows.

Incorrect Data Types

For example:

- Convert numerical columns stored as text into numeric format.
- Convert date columns into datetime format where applicable.

Inconsistent Values

Check categorical variables for:

- Incorrect spelling
- Different capitalization
- Unnecessary spaces
- Duplicate category names

Task 5: Analyze the Target Variable

Analyze the distribution of:

- Customers who churned
- Customers who did not churn

Calculate the percentage of each class.

This is important because customer churn datasets may have **class imbalance**.

Week 1 Deliverables

Submit:

- Dataset source or raw dataset
- Jupyter Notebook or Google Colab notebook
- Cleaned dataset

- Data dictionary
- Problem statement
- Data quality report
- Target variable distribution analysis

WEEK 2: Exploratory Data Analysis & Feature Engineering

Objective

Understand the factors associated with customer churn and prepare the dataset for Machine Learning.

Task 1: Perform Exploratory Data Analysis

Generate descriptive statistics for numerical variables.

Analyze:

- Customer tenure
- Monthly charges
- Total charges
- Customer demographics
- Contract type
- Payment method
- Subscription/service details

Task 2: Analyze Churn Patterns

Investigate churn across different customer groups.

Examples:

- Churn by contract type
- Churn by tenure
- Churn by payment method
- Churn by monthly charges
- Churn by internet/service type
- Churn by customer demographics

Create at least **8 meaningful visualizations**.

Task 3: Identify Key Churn Drivers

Answer questions such as:

- Which contract type has the highest churn?
- Do newer customers churn more frequently?
- Are customers with higher monthly charges more likely to churn?
- Does payment method appear related to churn?
- Which services are associated with higher churn?

Document at least **5 key insights**.

Task 4: Feature Engineering

Create or improve features where appropriate.

Examples:

- Tenure groups
- Number of subscribed services
- Total service indicators
- Spending categories
- Customer value groups

Interns must clearly document:

- The new feature
- How it was created
- Why it may be useful for predicting churn

Task 5: Feature Preprocessing

Prepare features for Machine Learning.

Tasks may include:

- Encoding categorical variables
- Scaling numerical variables where appropriate
- Removing irrelevant variables
- Handling outliers where justified
- Separating features from the target variable

Week 2 Deliverables

- EDA notebook
- Minimum of 8 visualizations
- Feature-engineered dataset
- List of engineered features
- At least 5 key insights
- Machine-learning-ready dataset

WEEK 3: Machine Learning Model Development

Objective

Build and compare multiple classification models.

Task 1: Prepare Training and Testing Data

Separate:

Features → **X**

Target → **Y**

Split the dataset into:

- Training dataset
- Testing dataset

Use a stratified train-test split where appropriate to maintain the churn class distribution.

Task 2: Build a Baseline Model

Start with:

Logistic Regression

Train a Logistic Regression model as the baseline.

Evaluate the model using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC

Task 3: Train Additional Models

Build and evaluate at least two additional classification models.

Task 4: Compare Model Performance

Create a model comparison table.

Example:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression					
Decision Tree					
Random Forest					
XGBoost					

Important Requirement

Do not choose the best model based only on **accuracy**.

Because churn may involve imbalanced classes, interns should pay close attention to:

- Precision
- Recall
- F1-score
- ROC-AUC

Week 3 Deliverables

- Training and testing datasets
- At least 3 trained classification models
- Model evaluation results
- Confusion matrices
- Model comparison table

- ROC curve, where applicable
- Initial model selection explanation

WEEK 4: Handling Class Imbalance & Model Optimization

Objective

Improve model performance and investigate the impact of class imbalance.

Task 1: Analyze Class Imbalance

Determine whether the dataset contains significantly more:

- Non-churn customers
- Churn customers

Explain why class imbalance can affect model performance.

Task 2: Experiment With Imbalance Handling

Where appropriate, experiment with techniques such as:

- Class weighting
- Oversampling
- Undersampling
- Model-specific class imbalance parameters

Compare results before and after applying the chosen approach.

Interns should document:

- The method used
- Why it was selected
- The impact on Precision
- The impact on Recall
- The impact on F1-score
- The impact on ROC-AUC

Task 3: Cross-Validation

Apply cross-validation to evaluate how consistently the model performs.

Compare the results across folds.

Task 4: Hyperparameter Tuning

Optimize the strongest models.

Recommended techniques:

- GridSearchCV
- RandomizedSearchCV

Tune models such as:

- Random Forest
- Decision Tree
- XGBoost, if used

Task 5: Final Model Selection

Compare:

- Baseline models
- Models after imbalance handling
- Optimized models

Select the best model based on both **technical performance and business usefulness**.

Week 4 Deliverables

- Class imbalance analysis
- Before-and-after model comparison
- Cross-validation results
- Hyperparameter tuning results
- Optimized model
- Final model selection report

WEEK 5: Model Interpretation, Business Insights & Final Presentation

Objective

Interpret the final Machine Learning model and translate the results into business recommendations.

Task 1: Final Model Evaluation

Evaluate the selected model using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC

Explain what each important metric means in the context of customer churn.

Task 2: Feature Importance Analysis

Identify the variables that most strongly influence churn.

Depending on the selected model, use techniques such as:

- Feature importance
- Logistic Regression coefficients
- Permutation importance

Analyze questions such as:

- What are the strongest predictors of churn?
- Which customer characteristics indicate high churn risk?
- Which services or contract types influence churn?

Task 3: Analyze Model Errors

Investigate:

False Positives

Customers predicted to churn who did not churn.

False Negatives

Customers predicted not to churn who actually churned.

Explain why these errors matter from a business perspective.

Task 4: Business Recommendations

Based on the analysis, provide at least **5 actionable recommendations**.

Recommendations must be based on actual findings from the dataset.

Possible areas include:

- Customer retention strategies
- Contract improvements
- Pricing strategies
- Customer engagement
- Loyalty programs
- High-risk customer intervention
- Service improvements

Task 5: Prepare Final Project

Create a complete, portfolio-ready project.

Final Notebook

Organize the notebook into:

1. Business Understanding
2. Data Collection
3. Data Understanding
4. Data Cleaning
5. Exploratory Data Analysis
6. Feature Engineering
7. Data Preprocessing
8. Model Development
9. Model Evaluation
10. Class Imbalance Handling

11. Hyperparameter Tuning
12. Final Model Selection
13. Feature Importance
14. Business Insights
15. Recommendations
16. Conclusion

Final Project Submission

Each intern should submit:

1. Complete Machine Learning Project

A properly organized Jupyter Notebook or Google Colab notebook.

2. Dataset

- Dataset source/link, or
- Cleaned and prepared dataset where permitted.

3. Exploratory Data Analysis

- Visualizations
- Key findings
- Churn analysis

4. Machine Learning Models

At least three classification models.

Recommended minimum:

- Logistic Regression
- Decision Tree or Random Forest
- Random Forest or XGBoost

5. Model Comparison

A comparison of all models using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

6. Feature Importance Analysis

Identify and explain the major factors contributing to customer churn.

7. README.md

The GitHub README should include:

Project Title
Project Overview
Business Problem
Dataset
Tools and Technologies
Project Workflow
Exploratory Data Analysis
Machine Learning Models
Evaluation Metrics
Key Findings
Business Recommendations
How to Run the Project
Project Structure

8. GitHub Repository

Suggested structure:

```
Customer-Churn-Prediction/  
├── data/  
│   ├── raw_data/  
│   └── processed_data/  
├── notebooks/  
│   └── customer_churn_analysis.ipynb  
├── src/  
│   └── preprocessing_and_modeling.py  
├── visuals/  
│   ├── churn_analysis.png  
│   ├── feature_importance.png  
│   └── model_performance.png  
├── README.md  
└── requirements.txt
```

9. Final Presentation or Demonstration

Prepare a **5–10 minute presentation** covering:

1. The business problem
2. The dataset
3. Data preparation
4. Key EDA findings
5. Feature engineering
6. Models tested
7. Model comparison
8. Final model
9. Feature importance
10. Business recommendations

Final Expected Outcome

By the end of the five-week internship, the intern should be able to complete an end-to-end Machine Learning workflow:

Customer Churn Business Problem → Data Preparation → EDA → Feature Engineering → Classification Models → Evaluation → Imbalance Handling → Optimization → Model Interpretation → Business Recommendations.